# VISVESVARAYA TECHNOLOGICAL UNIVERSITY

**Jnana Sangama, Belagavi – 590018**

*An Internship Report*

*On*

# "Full Stack Developer"

01/02/2026 to 01/04/2026

*Submitted in partial fulfillment of the requirements for the award of the degree of*

**Bachelor of Engineering**

**in**

**Electronics and Communication Engineering**

Submitted by

**KAMAL S 1VI22EC066**

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**VEMANA INSTITUTE OF TECHNOLOGY**

BENGALURU – 560034

**2025-2026**

**Karnataka ReddyJana Sangha**

# VEMANA INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi)
Koramangala, Bengaluru–560034

## DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

# CERTIFICATE

This is to certify that the internship entitled **"Full Stack Developer"** is a bonafide work carried out by **KAMAL S (1VI22EC066)** in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in **Electronics and Communication Engineering**, Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026.

It is certified that all corrections and suggestions indicated for internal assessment have been duly incorporated in the report. The internship report has been approved as it satisfies the academic requirements prescribed for the said degree.

| **Internal Supervisor** | **External Supervisor** | **HOD** | **Principal** |
|---|---|---|---|
| (Prof. Suma B V) | (Mr. Maneesh Kumar Varma) | (Dr. Suneeta) | (Dr. Vijayasimha Reddy B G) |

| **Name of the Examiners** | **Signature with Date** |
|---|---|
| 1. _____ | _____ |
| 2. _____ | _____ |

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I also acknowledge the support of Supram Technologies for providing structured learning resources and an environment that encouraged systematic understanding, practical application, and continuous improvement.

**Kamal S**

1VI22EC066

# ABSTRACT

During my internship at Supram Technologies as a Full Stack Developer, I gained exposure to multiple domains including industrial UI design, embedded systems, middleware, and system-level development.

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## Chapter 1

# INTRODUCTION

This chapter introduces the internship.

## 1.1 Internship Scope and Objectives

The scope was to gain practical knowledge.

## 1.2 Relevance of Full Stack Developer to Electronics and Communication Engineering

Full stack development is relevant to ECE.

## 1.3 Evolution of Practices in the Field

Engineering systems have evolved significantly.

## 1.4 Current Trends and Technologies

Current trends include embedded AI platforms.

## Chapter 2

# ORGANISATION PROFILE

Overview of Supram Technologies.

## 2.1 Introduction

Supram Technologies is a technology company.

## 2.2 Vision and Mission of the Organization

To be a leader in embedded systems.

## 2.3 Programs and Services Offered

Internship programs and training workshops.

## 2.4 Organizational Impact and Reach

Significant contributions to the field.

## 2.5 Core Values

Innovation, integrity, and excellence.

## Chapter 3

# WORK DONE / METHODOLOGY

Work completed during the internship.

## 3.1 Overview of Internship Work

Multiple projects spanning UI design and embedded systems.

## 3.2 Tools and Technologies Used

React, Node.js, Python, ROS.

## 3.3 Work Process and Methodology

Agile methodology with weekly sprints.

## 3.4 Key Project and Case Study

Building an industrial dashboard.

## 3.5 Challenges Faced

Integrating multiple systems.

## 3.6 Solutions Implemented

Using middleware for system integration.

## Chapter 4

# RESULTS AND DISCUSSION

Results and discussion.

## 4.1 Outcomes of the Work

Several working prototypes.

## 4.2 Learning Outcomes

Understanding of embedded systems.

## 4.3 Analysis

Valuable hands-on experience.

## Chapter 5

# CONCLUSION

Summary of the internship.

## 5.1 Summary

Comprehensive exposure to full stack development.

## 5.2 Personal Growth

Improved technical skills.

## 5.3 Future Scope

Deeper exploration of embedded AI.